

Homero Raúl Vargas Cruz

Staff Full-Stack Engineer — I design platforms, not one-off apps

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PROFILE

Fourteen years turning work businesses still do by hand into systems the people who need them can operate alone. I build the **platform** as readily as the product on top of it: a TypeScript framework of my own design carries twelve commercial desktop applications, along with the licensing server and metered gateway that sell them. Equally at home in Laravel/PHP, React/TypeScript, Node and Python, with LLM integration in production.

HOW I WORK

I build platforms, not one-off apps.

When a capability appears twice, it moves into a shared layer and everything else inherits it.

I replace the spreadsheet, not the button.

I match the layout people already know, because adoption depends on it feeling familiar.

I push configuration to the people who need it.

Rather than hardcode business rules, I build the editor. Rules change weekly; deploys should not.

I build guardrails before I need them.

Migrations that refuse to run destructively, validation that honours an abort, credentials that retire themselves.

EXPERIENCE

Founder & Principal Software Engineer — Innobox R&R

Jan 2018 — Present

- Designed **@innoboxrr/forg**e, a TypeScript framework for offline-first, licence-sold Electron desktop applications: 20 modules spanning kernel, migrations, job queue, event bus, auth adapters, an embedded Fastify server, signed auto-updates and an MCP server.
- Made backends adapter-swappable, so the same product runs on local SQLite, Supabase or PostgreSQL without a rewrite. **Twelve commercial products ship on it**, each with its own versioned release channel.
- Built the monetisation layer the catalogue depends on: a multi-product licensing server on Supabase Edge Functions (Deno), and a gateway fronting object storage and AI inference with entitlement checks and credit metering.
- Published **40+ open-source packages** to Packagist and npm, every one extracted from delivery work rather than written speculatively.
- Delivered enterprise platforms for clients: SeguroPro (insurance CRM), iTec School, SiConsulta, ProfeMX.

Backend Engineer — ActualSales

Dec 2023 — Present

- Maintain the shared internal library the platform's services depend on, and the services that consume it.
- Built rule builders operations staff drive from the browser and test against real data before committing, removing a deploy from every routing and onboarding change.
- Replaced a manual Excel reporting workflow with a period-comparison pivot builder: day-over-day deltas, expandable drill-downs and spreadsheet export.
- Added operational guardrails across the platform and replaced a charting dependency with a server-rendered SVG renderer when the library could not produce the required layout.
- Keep third-party integrations current, including major API version migrations.

Full-Stack Developer — eScire

Jul 2023 — Jan 2024

- Led a learning-management platform for **Tecnológico de Monterrey**, configuring learning environments for master's and doctoral cohorts.
- Integrated research APIs and built statistical-analysis pipelines over publication data.

- **Founded the school's Technology Department and led a team of 8.**
- Built the systems the institution runs on — enrolment, finance, social service, procedures and R&D — from first design through years of iteration.

- Built a multi-tenant education platform letting teachers publish and sell courses, with encrypted video delivery and integrated payments.
- Teach what I build: **25 open teaching repositories** at Laravelers Academy.

SELECTED PROJECTS

Caso Culpable [casoculpable.com](#) · [live](#) — deductive crime-fiction game built as a hexagonal monorepo: a zero-dependency pure domain and a **deterministic solver that proves every case is logically solvable before it can be published**. Architecture and secret-scanning gates run in CI.

LeadOps [Electron](#) · [Claude SDK](#) — prospecting workstation with LLM-powered enrichment, headless-browser collection and geospatial mapping.

Chispa [offline-first](#) — children's education platform where the pedagogy engine is kept separate from the shell: per-node mastery modelling, spaced repetition and an adaptive session planner. A child's progress never leaves the machine.

xtts_local [Python](#) · [PyTorch](#) — voice cloning (Coqui XTTS) packaged with PyInstaller and shipped as a licensed Windows application. Machine learning delivered as a product, not a notebook.

Tipster [React Native](#) · [Expo](#) — Laravel API with a production mobile client: in-app purchases, push notifications, OAuth and full localisation.

TECHNICAL SKILLS

Languages	TypeScript · JavaScript · PHP · Python · SQL · Bash
Frontend & Desktop	React · Electron · Vue.js · React Native (Expo) · Vite · Tailwind
Backend	Node.js · Fastify · Laravel · Express · Deno edge functions · Livewire
Data	PostgreSQL · MySQL · SQLite · Supabase
AI / ML	Anthropic Claude SDK in production · Model Context Protocol (MCP) · PyTorch · Coqui XTTS
Cloud & DevOps	AWS (EC2, S3, Lambda, IAM, Route 53, SES, CloudWatch, CodeDeploy) · Docker · Kubernetes · GitHub Actions · GitLab CI · Sentry
Practices	Hexagonal architecture · offline-first & sync · monorepos · migrations as code · multi-tenancy · signed releases & delta updates · REST/SOAP · i18n · accessibility

EDUCATION & LANGUAGES

M.Sc. Innovation & Educational Technology — Universidad del Valle de México (GPA 10/10)	2015 — 2018
B.Eng. Computer Systems Engineering — UTEL University (GPA 9.4/10)	2014 — 2018
B.A. Psychology — Instituto Universitario del Estado de México	2011 — 2015
Languages — Spanish (native) · English (advanced)	